

1529. Kth Missing Positive Number:

Given an array `arr` of positive integers sorted in a **strictly increasing order**, and an integer `k`.

Return the `kth` **positive** integer that is **missing** from this array.

Example 1:

Input: `arr = [2,3,4,7,11]`, `k = 5`

Output: 9

Explanation: The missing positive integers are `[1,5,6,8,9,10,12,13,...]`. The 5th missing positive integer is 9.

Example 2:

Input: `arr = [1,2,3,4]`, `k = 2`

Output: 6

Explanation: The missing positive integers are `[5,6,7,...]`. The 2nd missing positive integer is 6.

Constraints:

- `1 <= arr.length <= 1000`
- `1 <= arr[i] <= 1000`
- `1 <= k <= 1000`
- `arr[i] < arr[j]` for `1 <= i < j <= arr.length`

Follow up:

Could you solve this problem in less than $O(n)$ complexity?

Solution: Time = $O(\log n)$ and Space = $O(1)$

```
class Solution:
    def findKthPositive(self, arr: List[int], k: int) -> int:
```

```
left, right = 0 , len(arr) -1
while left <= right:
    mid = (left + right) // 2
    # Formula for finding missing numbers
    missing = arr[mid] - (mid+1)

    if missing < k:
        left = mid + 1
    else:
        right = mid - 1

return left + k
```